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VANDERBILT
2026

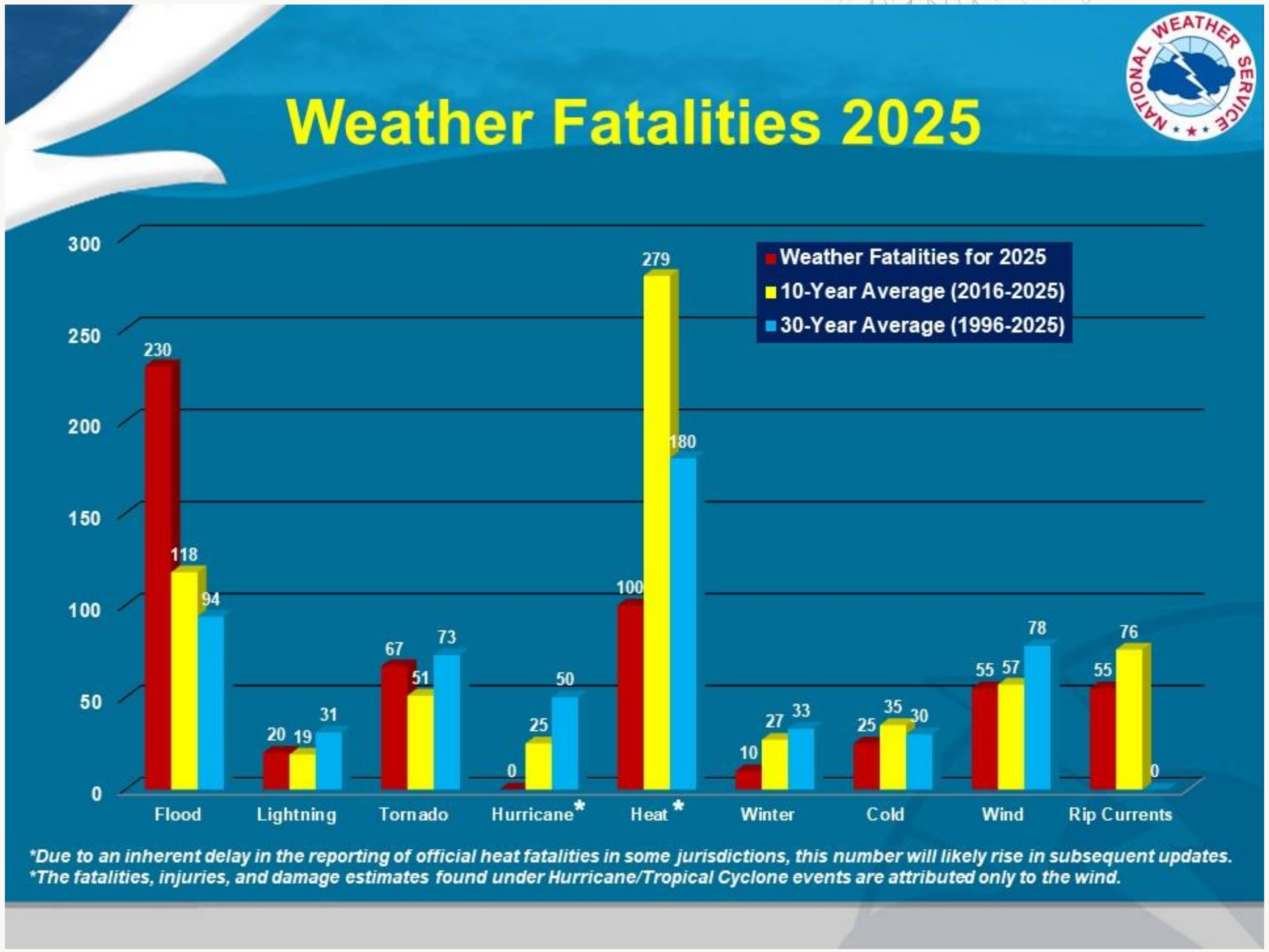
Sustainability in Scientific Research: Challenges and Opportunities

J. Magnolia Gilligan

Sept. 5, 2026

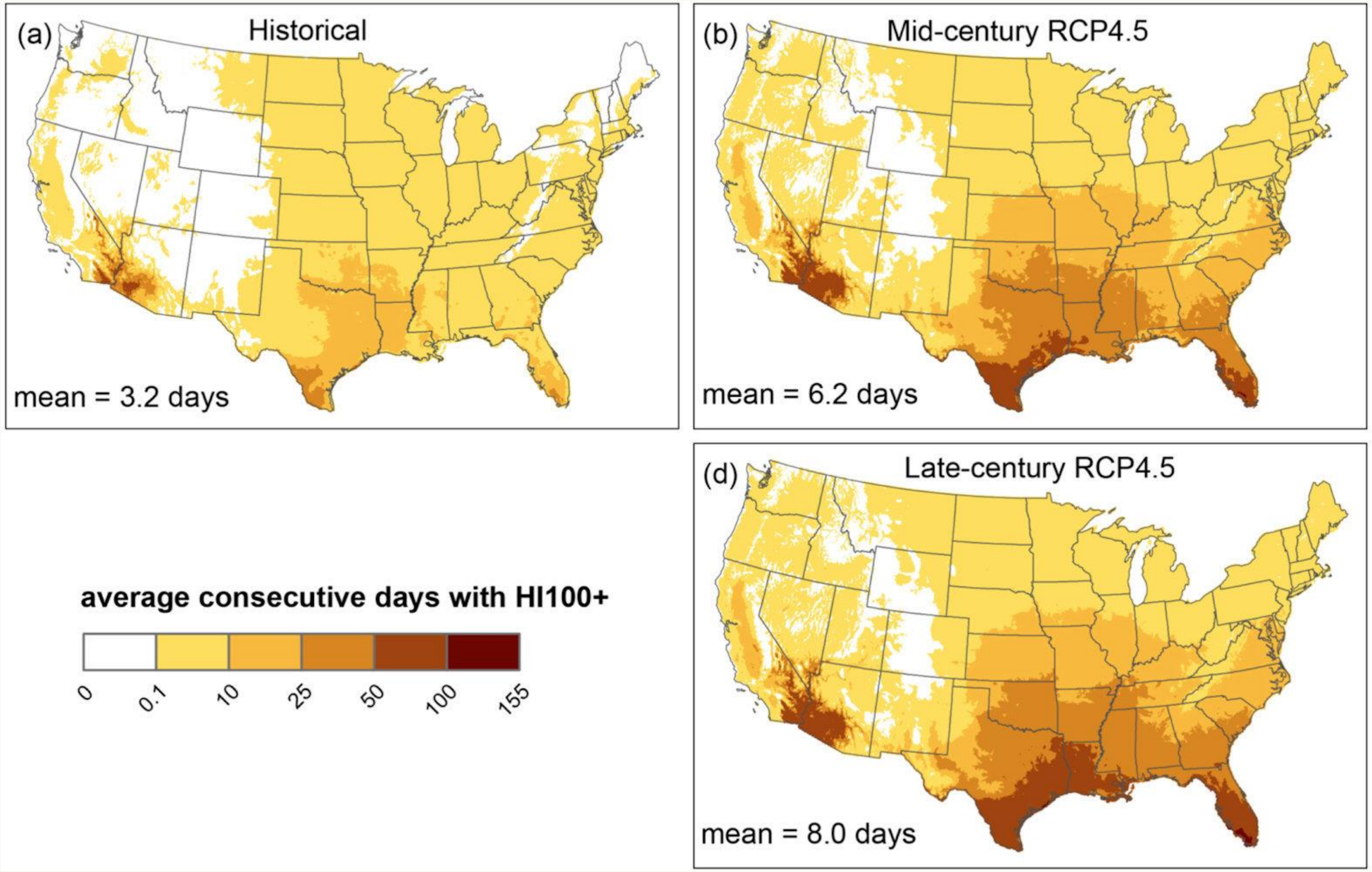
Climate Change and Health

- Extreme heat waves are now 50 times more frequent than they used to be
- 6 of the 10 deadliest heat waves in history happened since 2000
  - 2003 European heat wave: 70,000 deaths
  - 2010 Russian heat wave: 50,000 deaths
- 2021 heat wave in Pacific Northwest killed more than 1200 in Washington, Oregon, Canada
- Since 2000, U.S. heat-waves caused additional 17,600 deaths per year among adults aged 65+.
  - Black people: 3 × mortality of white people
  - High-poverty neighborhoods: 2 × mortality
  - Abundant green-space: dramatically lower risk
  - J.P. Healy et al. Lancet Planetary Health 10, 101432 (2026) doi: 10.1016/j.lanplh.2026.101432
- Extreme Heat is the leading cause of weather-related fatalities in U.S.



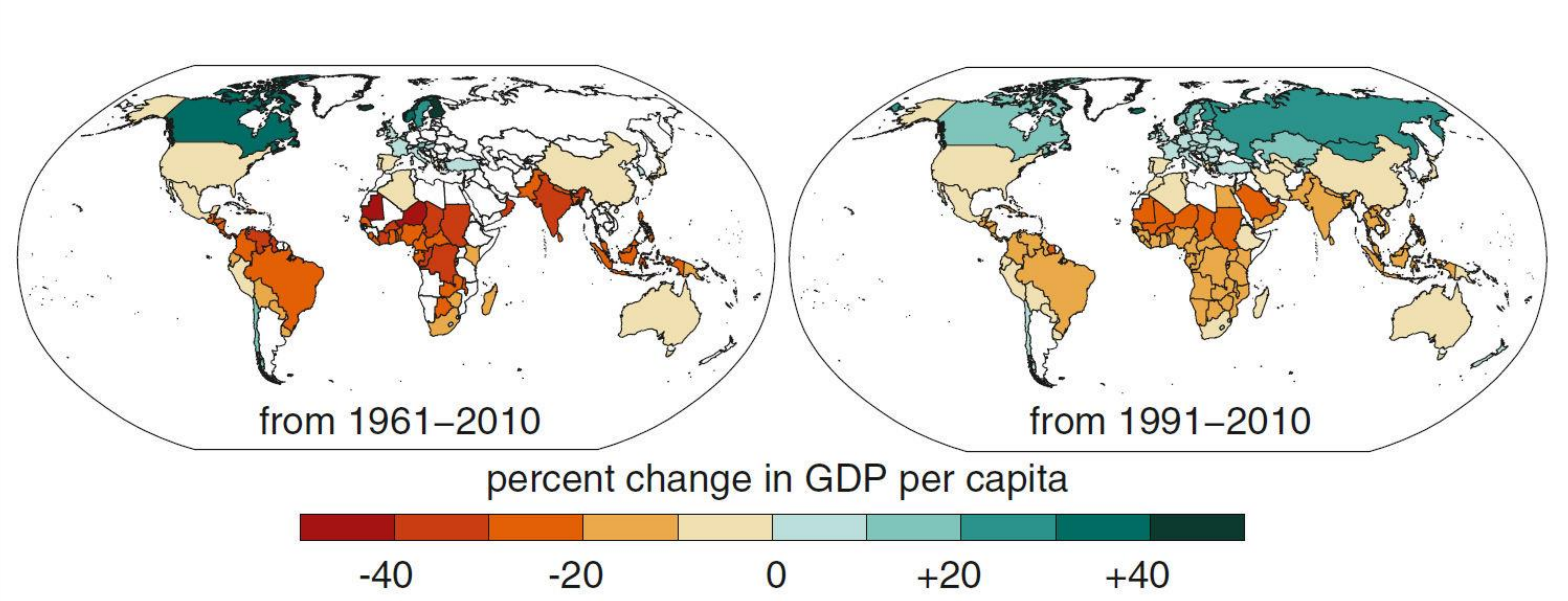
National Weather Service

Future Heat in U.S.



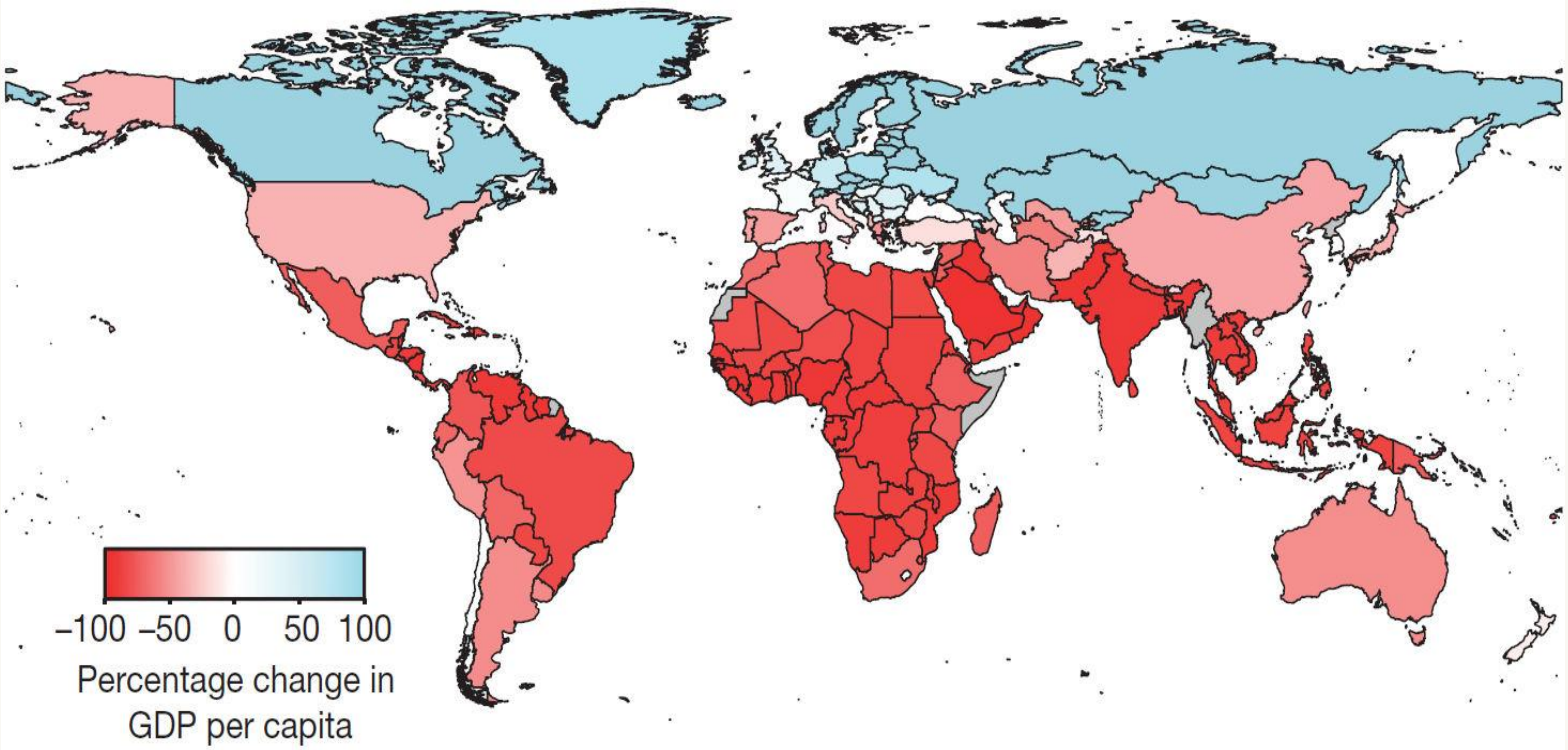
K. Dahl *et al.*Environ. Res. Commun. 1, 075002 (2019)
doi: 10.1088/2515-7620/ab27cf

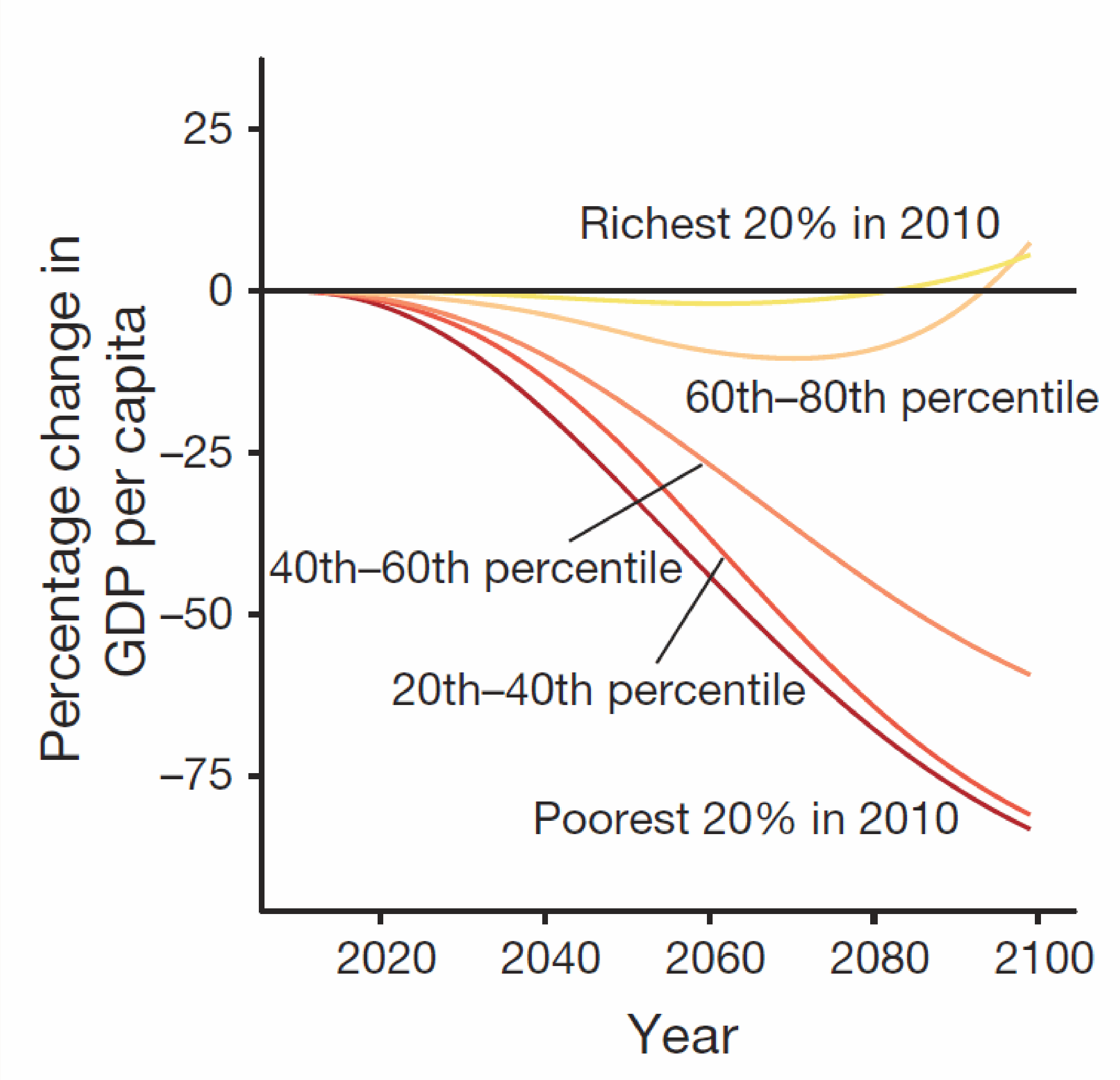
Global Warming Is Already Affecting Economic Inequality



N.S. Diffenbaugh & M. Burke, Proc. Nat'l Acad. Sci. 116, 9808 (2019) 10.1073/pnas.1816020116

Future World Economy under Extreme Warming





M. Burke, S. Hsiang, & E. Miguel, Nature 527, 235 (2015) doi: 10.1038/nature15725

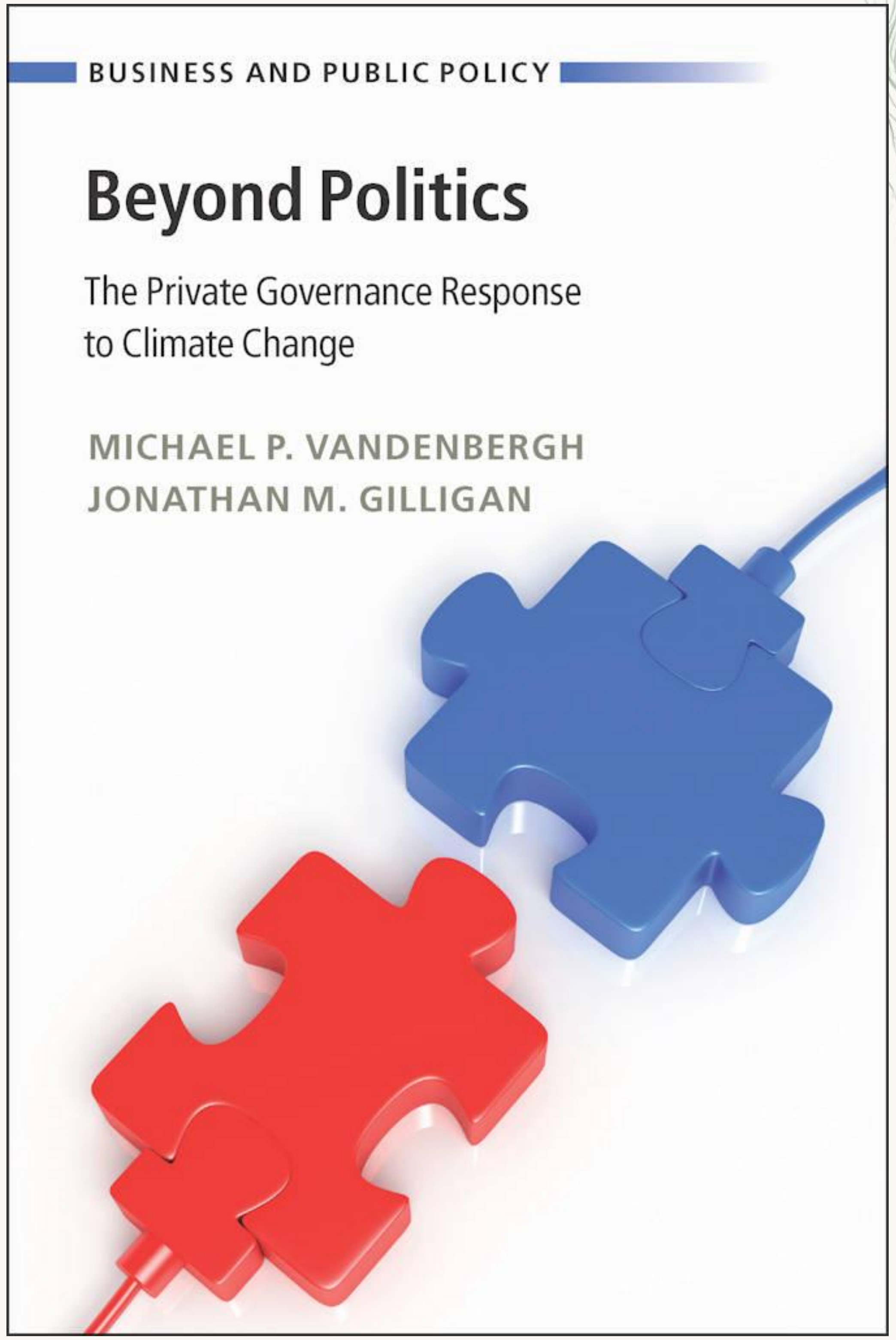
What Can We Do?

- Public Governance
  - Command & Control
    - Every car must have a catalytic converter
    - CAFE standards for fuel economy
  - Market-based regulations
    - Ronald Coase (1960)
    - C. Boyden Gray (1980s)
    - Tradeable permits
    - Emissions Taxes

- Political Opportunity Cost
  - Market-based regulations are economically optimal, once enacted
  - But after 40 years of trying we still don't have the political will to price GHG emissions
  - "A good solution applied with vigor now is better than a perfect solution applied 10 minutes later"
    — George S. Patton (attrib)
  - Gilligan & Vandenbergh, *Accounting for Political Feasibility in Climate Instrument Choice 32 VA ENVT'L L. J. 1–26 (2014)*

What Can We Do?

- Private actors are not as powerful as governments, but can move faster
  - It’s not just what you do yourself.
    It’s also how you influence others.
  - Businesses:
    - Walmart, etc.
    - Supply-chain contracting as coercive private governance
  - Private households:
    - Behavioral Wedge (PNAS 2009)
    - Social norms and influence
  - Employee pressure on employers
- Estimated impact:
  - Up to 1 billion tons CO_2e per year in US



Supply-Chain Contracting

- 73% of the largest corporate buyers expect to reject suppliers based on environmental performance
- 200 large corporate buyers are requiring 24,000 of their suppliers to cut greenhouse gas emissions
  - 1.8 billion tons cut in 2021 alone
- US Inflation Reduction Act would have taken until 2030 to cut 1 billion tons per year

- Question: Medical centers & research labs are large corporate purchasers:
  - How can your lab and/or clinic emphasize sustainable purchasing?
  - How can you pressure your suppliers to become more sustainable?

Los Angeles Times

OPINION

Op-Ed: Here’s how companies can strong-arm their suppliers into cutting carbon emissions

A Tyson Foods pork processing plant in Indiana. The Arkansas-based company recently faced pressure from Walmart, one of its major distributors, to curb emissions. Tyson achieved that by pushing its own suppliers to rein in greenhouse gases. (Michael Conroy / Associated Press)

BY MICHAEL P. VANDENBERGH AND JONATHAN M. GILLIGAN

OCT. 4, 2022 3 AM PT

The Things We Don't Notice that Matter

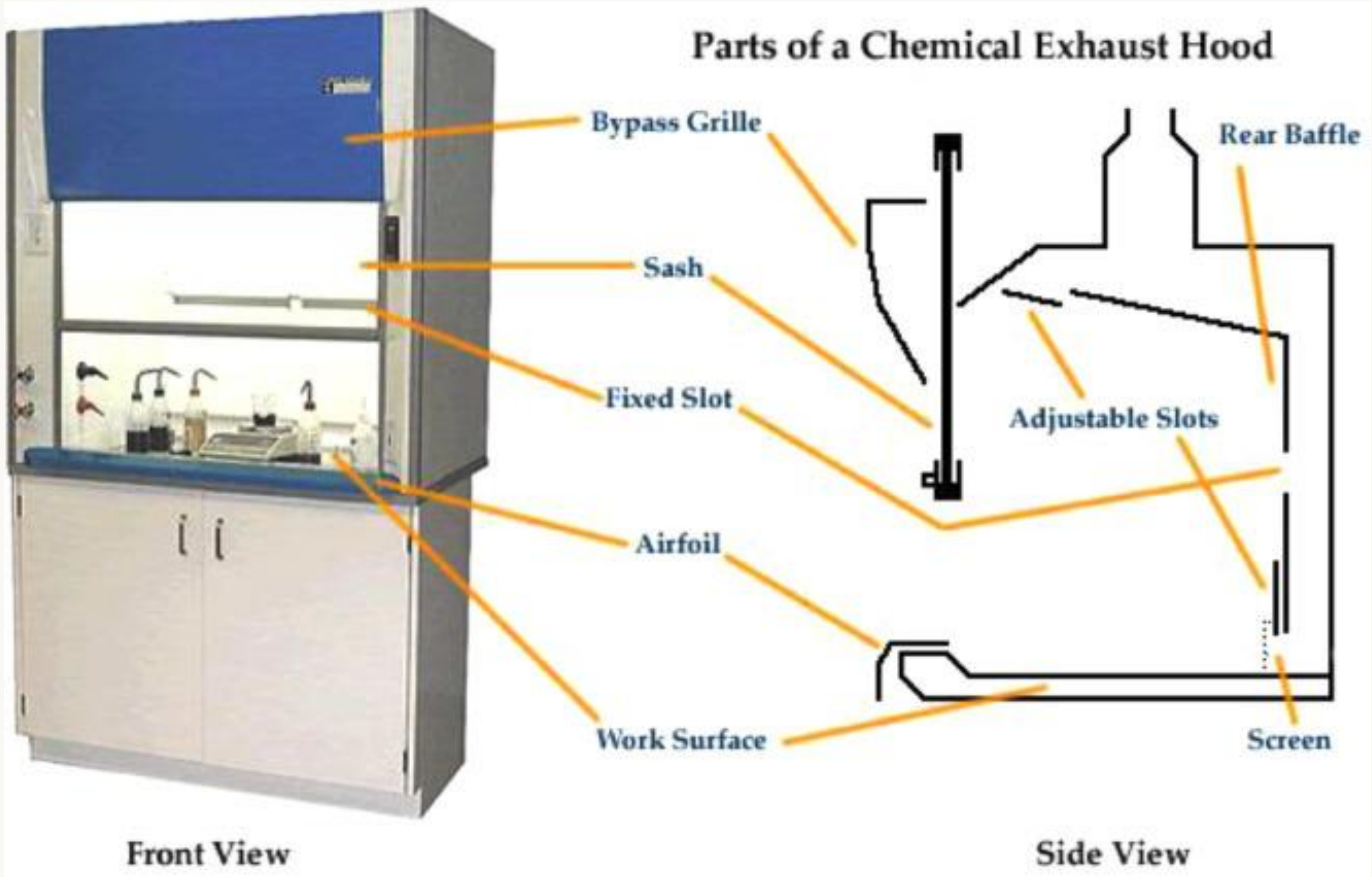
- Businesses often find they save money, increase profits when they reduce emissions
  - Walkers Crisps saved more than $500,000 wit a 50% reduction in GHG emissions from growing and processing potatoes.
- Why don't businesses find these savings sooner?

Virgin Airlines

- Cut 21,000 tons of CO_2 emissions
  - Saved $250 per ton of avoided emissions
  - Total savings: $4.2 million
- Pilots reported greater job satisfaction

Laboratory Fume Hoods

- 500,000 to 1,500,000 fume hoods in US labs
- Consume 26 TWh per year
  - Energy cost: $4.2 billion/year
- Energy savings:
  - Design & engineering:
    - Reduce number & size of hoods
    - Improved sash design
    - Automated sash-closure systems
    - High-efficiency ventilation systems
  - Behavior
    - Develop habits to close sash
- PG&E, LBL (2007):
  - 75% energy reduction is possible
- Harvard (2005):
  - 30% energy reduction from user behavior
  - $240,000, 300 tons greenhouse gases



A. Gillis *et al.* Curr. Res. Ecol. & Social Psych 3, 100063 (2022)
doi: 10.1016/j.cresp.2022.100063

Vanderbilt Experiment

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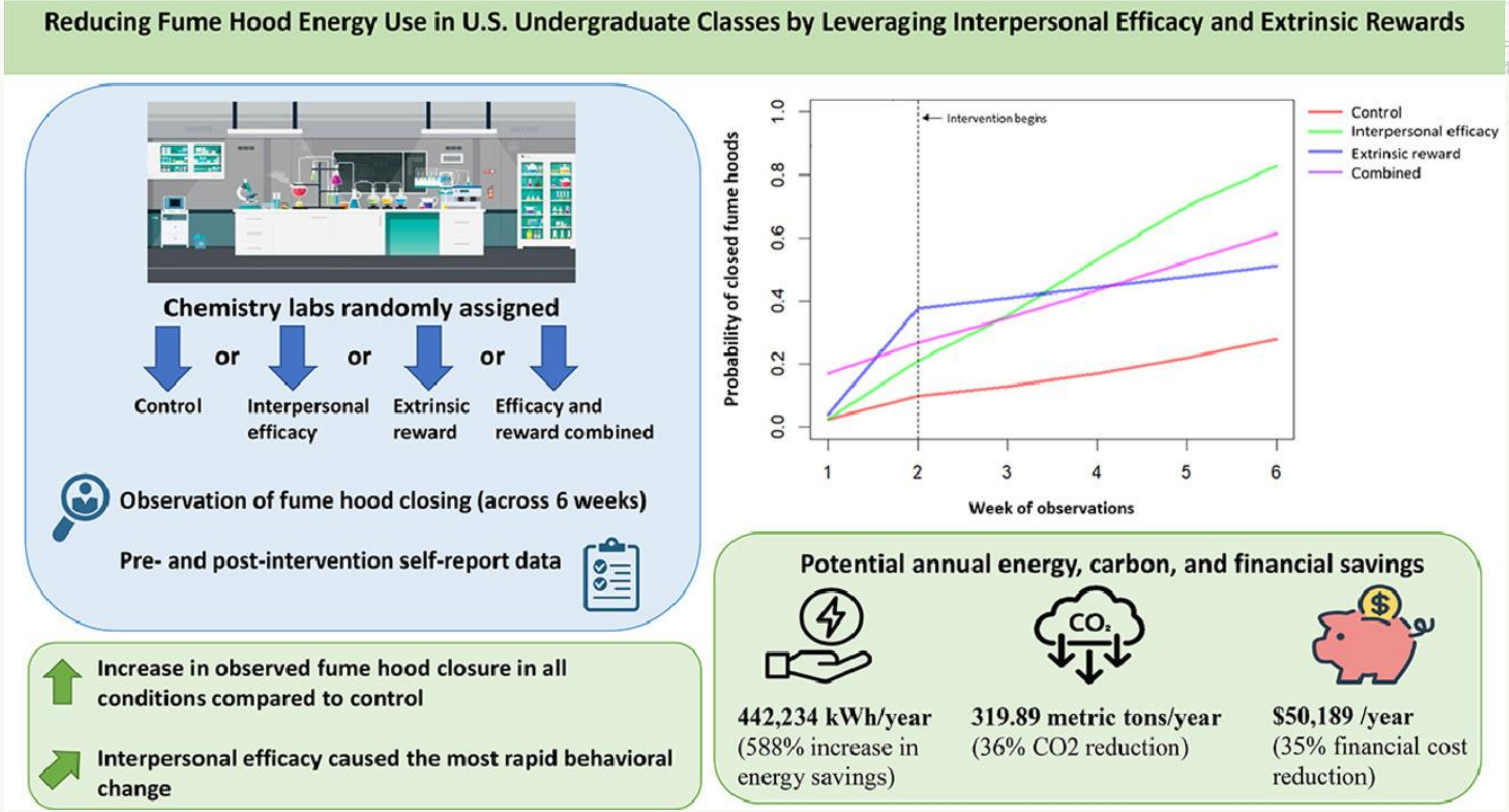

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doi: 10.1016/j.cresp.2022.100063

brainhack-vandy.github.io

Opportunities to Reduce Emissions in Your Labs

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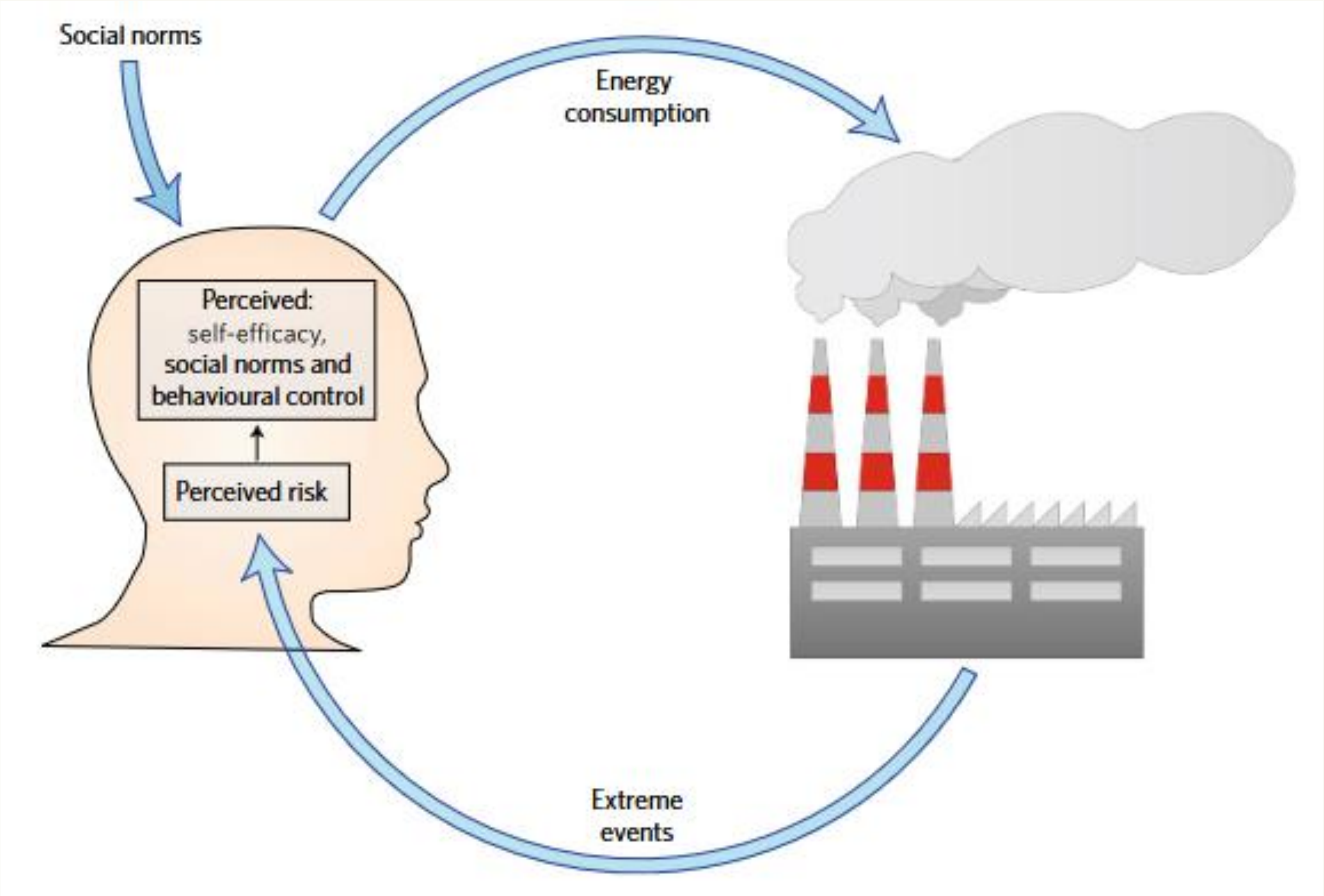
Top GHG Emissions Sources at U.S. Hospitals

- Electricity Generation and Use
- Staff Commuting
- Medical Device, Supplies & Pharmaceutical Procurement
- Work-Related Air Travel
- Food Procurement
- Anesthetic Gas Use
- Waste Disposal
- Fleet Vehicles
- Refrigerant Releases
- Emergency Generators

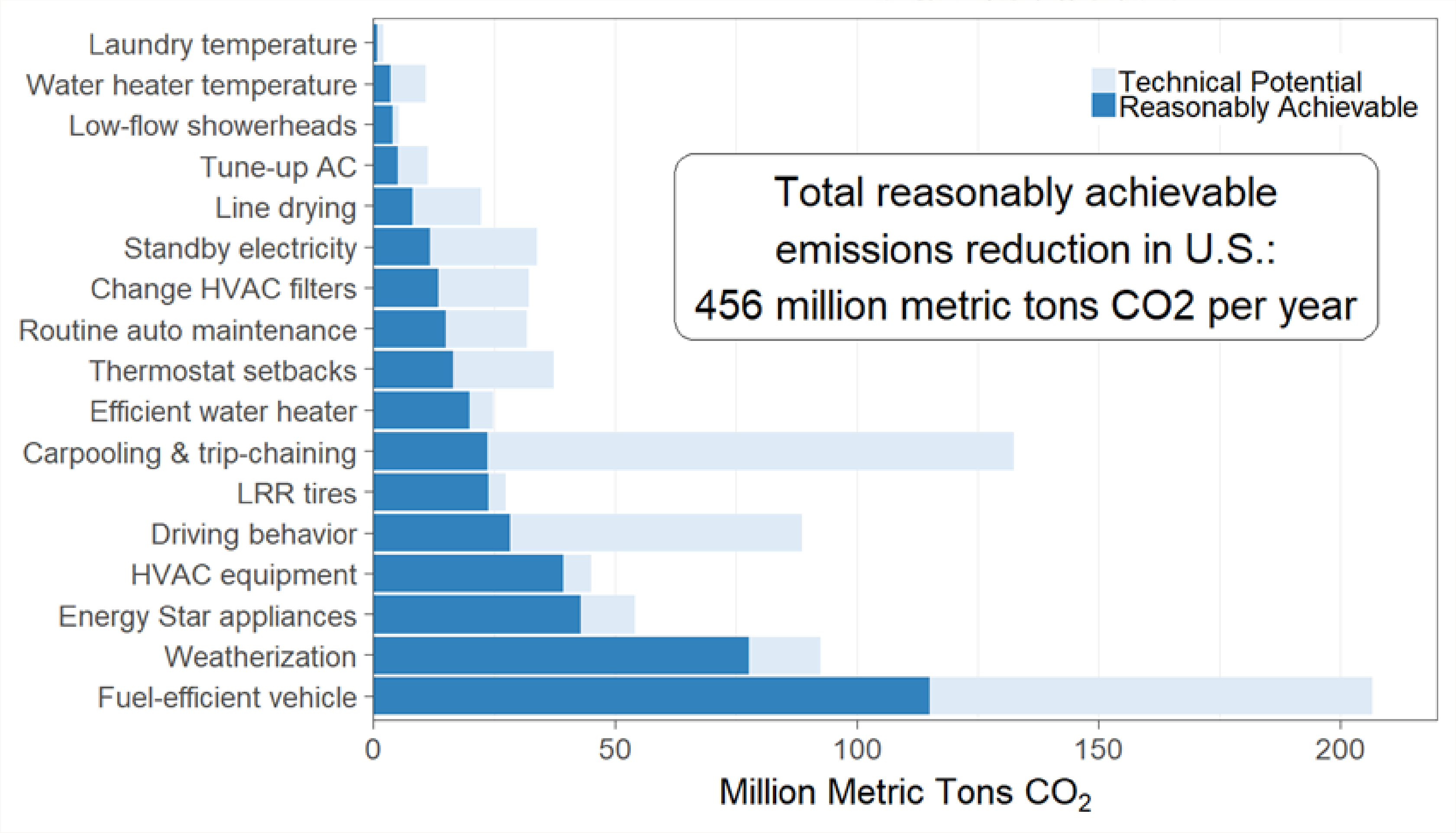
- Where are opportunities to reduce emissions in your labs?
- What obstacles are there?
- How might you overcome the obstacles?

Behavioral Research and Sustainability

- Emissions are driven by human behavior
  - Changing behavior can significantly reduce emissions
  - Understanding drivers of behavior, decision-making are crucial
- Individual decision-making
- Collective behavior



J.M. Gilligan, Nature Climate Change 8, 14 (2018)
doi: 10.1038/s41558-017-0038-0



PNAS

Household actions can provide a behavioral wedge to rapidly reduce U.S. carbon emissions

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Edited by Elinor Ostrom, Indiana University, Bloomington, IN, and approved September 11, 2009 (received for review August 2, 2009)

Most climate change policy attention has been addressed to long-term options, such as inducing new, low-carbon energy tech-

available technologies in U.S. homes and nonbusiness travel by means of behaviorally oriented policies and interventions. This